

RESEARCH ARTICLE

Application of IRSA-BP neural network in diagnosing diabetes

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Abstract

Within the healthcare sector, the application of machine learning is gaining prominence, notably enhancing the efficiency and precision of diagnostic procedures. This study focuses on this key area of diabetes prediction and aims to develop an innovative prediction method. Using the data set published by Kare, this paper constructs and compares various intelligent systems based on multilayer algorithms, and specifically introduces improved reptile search algorithm (IRSA) to optimize the weight and threshold initialization of traditional backpropagation (BP) neural networks. This improvement aims to improve the network performance and accuracy in diabetes detection. In the study, the IRSA-BP hybrid algorithm and many other machine learning algorithms were used for diabetes prediction, and the algorithm performance was comprehensively evaluated using multiple classification metrics. The experimental results showed that the IRSA-BP algorithm performed the best among all the evaluated algorithms, with an accuracy of up to 83.6%, showing its superior performance in diabetes prediction. Therefore, the IRSA-BP classifier has an important potential for application in the medical field. It can assist medical professionals to identify diabetes risk earlier and assess the condition more accurately, thus improving diagnostic efficiency and accuracy. This is important for early intervention and treatment of patients with diabetes and to improve their health status and quality of life.

1. Introduction

Machine learning, a prominent application within the sphere of artificial intelligence, is experiencing swift growth, notably in the medical field. This technology is particularly significant in two key areas: diagnosing diseases and overseeing the administration of treatments [1,2]. Many researchers are exploring how to improve the timeliness and accuracy of medical diagnosis through ML [3]. It can be used for disease prediction, feature selection, etc., as described in [4], feature selection can significantly improve the accuracy of classification models by filtering out the most influential

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subset of features. In the healthcare system, an accurate diagnosis is critical. According to the World Health Organization, more than 138 million patients are harmed by medical errors every year, with 2.6 million killed [5]. This situation not only harms patients, but also leads to a waste of medical resources, such as performing unnecessary diagnostic tests [6]. Moreover, diagnostic errors may also increase healthcare costs and reduce public trust in the healthcare system.

Physician time to enter and analyze data on computers increases and, correspondingly, time to communicate with patients decreases [7]. Recently, Artificial-Intelligence (AI) tools based on ML algorithms have become important aids in the field of medical diagnosis. These algorithms enable in-depth analysis of huge medical data and provide predictions, providing unprecedented support to the healthcare industry. For example, in the prediction of glaucoma, scholars have achieved remarkable results [8].

Data mining technologies are able to translate large amounts of raw medical data into valuable information, helping medical decision makers to make more informed decisions and predictions [9]. By analyzing this information, ML can provide decision support and predictive analysis. These predictions can reveal key health risks in advance, allowing patients and healthcare providers to take timely preventive measures, with the potential to completely avoid disease development [10].

ML holds the promise of transforming the healthcare sector through the delivery of expedited and precise diagnostic predictions. With the continuous progress of telemedicine technology, the ML method can bring better medical services to the areas with poor medical resources around the world. Moreover, ML technology helps to reduce unnecessary diagnostic tests by improving the accuracy of diagnosis and treatment, thereby reducing overall healthcare costs and potentially increasing the time for healthcare providers to communicate face-to-face with patients.

ML tools, in comparison to conventional methodologies, markedly enhance numerous evaluative metrics and augment the swiftness and neutrality of data analysis [11]. Specifically, the prevalence of certain complex diseases may be significantly diminished by implementing early preventative strategies. Characterized by elevated blood glucose levels, diabetes mellitus is a chronic metabolic condition [12] and a predominant factor in global disease burden and death rates, earning its classification as a worldwide health epidemic [13]. The term 'diabetes mellitus' encompasses a spectrum of disorders related to glucose regulation, with the primary classifications being type 1, type 2, gestational, and other distinct subtypes of diabetes [14–17]. These diseases may lead to multiple complications such as cardiovascular disease, renal lesions, decreased vision loss, foot problems and neurological injury like [18]. According to the US Centers for Disease Control and Prevention (CDC), diabetes is a primary contributor to death rates among both genders in the United States. The CDC and the New York State Department of Health have documented key facts regarding diabetes: 1) In the United States, one person dies every 54 seconds due to diabetes-related complications. 2) About 80,000 people die from diabetes in the United States, accounting for one-tenth of all deaths. 3) For individuals of both male and female genders, diabetes is a major cause of death. 4) Diabetes mellitus type

2 represents the predominant form of diabetes, constituting over 90% of diagnosed cases. 5) Approximately 253,000 Americans are diagnosed with diabetes every year. Of these, 157,000 were first diagnosed, while 96,000 were those who already had pre-diabetes. These data highlight the serious threat to public health of diabetes, highlighting the urgent need for improved strategies for diabetes prevention, diagnosis and treatment.

The global cost of treating diabetes alone was about \$76 billion in 2020 and is expected to exceed \$1.7 trillion in [19] by 2040. As a result, the forecasting and prevention of diabetes emerges as one of the more and more important subjects in clinical data analysis. However, due to multiple risk factors such as obesity, high cholesterol, and hypertension, it is arduous to compute the chance of suffering from diabetes by manual means [20]. Fortunately, ML enables the prediction of numerous diseases through the analysis of vast medical datasets. Scholars have introduced various ML algorithms designed to dissect extensive and intricate medical data, thereby facilitating the forecasting of diabetes by medical practitioners.

Multiple ML models were compared in [21], including Glmnet, Random Forest (RF), XGBoost, LightGBM, etc., to predict undiagnosed type 2 diabetes mellitus (T2DM). Scientists have devised a diabetes forecasting methodology that integrates SMOTE oversampling with RUS undersampling to tackle data imbalance, and utilizes Optuna for the automated hyperparameter optimization of the LightGBM model, thereby bolstering the model's predictive precision [22]. Furthermore, they have explored diabetes detection through the application of both machine learning and deep learning approaches, assessing various models including support vector machines (SVM), random forests (RF), logistic regression (LR), gradient boosting machines (GBM), and neural networks [23]. A method relying on physical activity measurements from mobile devices for detecting type 1 diabetes is presented in the [24]. Physical activity data collected by mobile devices were analyzed using machine learning algorithms to predict whether an individual has type 1 diabetes. Furthermore, this study employed an artificial neural network to forecast diabetes and developed a predictive model grounded in neural network principles, which was subsequently validated through experimental assessments. In addition, this research delves into methods to boost the model's predictive accuracy by refining the architecture and tuning the hyperparameters of the neural network [25]. A support vector machine (SVM) to predict the diagnosis of diabetes is achieved in the [26]. Studies suggest that the diagnosis of diabetes involves multiple factors and is susceptible to human error. The research reported in [27] has developed a computational framework that employs machine learning algorithms for predicting and clinically diagnosing diabetes. The underlying hypothesis of this study posits that optimizing feature selection and addressing missing data through imputation techniques could potentially enhance the efficacy of classification algorithms in predicting diabetes.

This paper presents an IRSA-BP network for diabetes prediction. The IRSA algorithm is an effective optimization technique to guide the optimization of BP neural network training, including determining the best weight and bias values. Following a series of experiments, the BP algorithm optimized by the IRSA method demonstrated superior performance when compared with alternative heuristic optimization techniques. The IRSA method is characterized by its user-friendliness, computational efficiency, and robustness with respect to control parameter settings. This paper Use famous performance evaluation indicators to compare the performance of IRSA-BP with MPA-optimized BP, RSA and AO. Datasets from the publicly accessible Kare repository served as benchmarks for evaluating the efficacy of various algorithms. These datasets comprise eight predictive features and a binary target variable, where the values 0 and 1 indicate the absence and presence of diabetes, respectively. The objective was to employ a suite of machine learning techniques, including the newly proposed IRSA-BP, to forecast the likelihood of an individual developing diabetes (1) or not (0), utilizing the eight medical attributes provided in the Kare dataset. The methodology presented in this study is designed to predict the onset of diabetes, thereby equipping patients and medical professionals with critical data to inform personalized treatment choices. Proactively implementing preventive measures can significantly improve patients' health outcomes.

In recent years, numerous studies have explored the application of machine learning algorithms for diabetes prediction, each employing different methodologies and achieving varying levels of accuracy. For instance, Khanam and Masoodi [28]

used machine learning techniques to achieve early detection and risk prediction of type 2 diabetes. The study employed multiple machine learning models, including K-Nearest Neighbor (K-NN), Bernoulli Naive Bayes (BNB), and Decision Tree (DT). Experimental results showed that the K-NN model performed best in detecting diabetes, with an accuracy of 79.6%, while the BNB model had an accuracy of 77.2%. The performance of the other models was relatively lower. In contrast, our proposed IRSA-BP model achieves a significantly higher accuracy of 83.6%, outperforming these existing approaches. This improvement can be attributed to the enhanced optimization capabilities of the IRSA algorithm, which effectively refines the weight and bias initialization of the BP neural network, leading to better convergence and predictive performance. Furthermore, our model demonstrates superior performance across multiple evaluation metrics, including AUC, precision, recall, and F1 score, compared to other state-of-the-art methods such as RSA-BP, MPA-BP, and AO-BP. These results underscore the potential of the IRSA-BP model in advancing the field of diabetes prediction, particularly in handling complex and high-dimensional medical datasets.

The layout of the ensuing sections within this manuscript is outlined as follows: Section 2 elucidates the methodologies and materials applied in our investigation. Section 3 details the experimental procedures and the outcomes they yielded. Section 4 encapsulates our findings and proposes future research avenues.

2. Materials and methods

2.1. Dataset

The dataset used in this study is provided by Khare. This dataset was originally from the National Institute of Diabetes, Digestive and Kidney Diseases (NIDDK) of the United States, and the quality of this dataset is high; The variables are rich and diverse, reflecting the patient's physical condition from multiple dimensions, providing comprehensive information for the diagnosis and prediction of diabetes, and can be freely used with CC0 public domain permission. Characterized as an imbalanced classification dataset, it comprises 769 samples, each with 8 predictive features and a binary target variable, with a particular focus on female subjects aged 21 and above, as detailed in Table 1.

This dataset is an openly available dataset. During its use, we strictly adhered to the data provider's usage terms and removed any remaining identifying information to prevent reidentification. As the data collection adhered to ethical guidelines, this usage has no direct impact on participants, so additional ethical approval is not required. Overall, the current data collection and processing methods comply with ethical standards.

To ensure dataset representativeness, we analyzed its demographic and clinical traits. The dataset focuses on women aged 21 +, which is relevant for diabetes prediction due to age – related and gender – specific factors like pregnancy. It has diverse values for key diabetes – related features.

Table 1. Description of the dataset features.

Features	Meaning
Pregnancies	The number of pregnancies is shown in units (units)
Glucose	Glucose levels in blood were expressed in milligrams per deciliter (units)
Blood Pressure	Blood pressure measurements are expressed in mmHg (units)
Skin Thickness	Represents the thickness of the skin
Insulin	Expressed the insulin levels in the blood
BMI	Meaning body mass index
Diabetes Pedigree Function	Represents the percentage of diabetes mellitus
Age	Represents the age of the individual.
Diabetes diagnosis	0 indicates diabetes.
	1 is expressed as having no diabetes mellitus.

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Although limited to females, this is justifiable considering gestational diabetes and pregnancy – related diabetes risk. Future studies could include males for broader generalizability. The dataset's wide age and feature range captures population variability.

We also compared the dataset with large – scale epidemiological studies. The similar distribution of key features like glucose levels, BMI, and age indicates that it's a reasonable representation of the diabetes – at – risk population.

The selection of the above predictive characteristics is based on a combination of diabetes-related factors. For example, pregnancies may be closely associated with the risk of gestational diabetes, and by analyzing this feature, a better understanding of the incidence of diabetes in the female population can be obtained. Glucose is one of the most important indicators for the diagnosis of diabetes, which directly reflects the metabolism of blood glucose in the body. Blood pressure, skin thickness, blood pressure, skin thickness, insulin and other characteristics are also associated with the onset and development of diabetes. The selection of these features aims to comprehensively cover the physiological indicators related to diabetes mellitus and provide rich information for the prediction of the model.

2.1.1. Exploratory data analysis. Exploratory data analysis (EDA) constitutes an essential phase in the initial examination of datasets. Its main function is to explore the patterns, associations between variables and outliers in the data, test the hypothesis, and verify the hypothesis [29] with the help of summary statistics and graphical presentation. As an important step in optimizing and adjusting a given dataset in various forms of analysis, EDA is committed to grasping the core characteristics of various entities in the dataset and drawing a clear picture of the characteristics and their interrelations. As depicted in Fig 1, the correlation matrix is visually represented as a tabular format that illustrates the correlation coefficients among two or more variables.

To determine Pearson's correlation coefficient, it is essential to analyze the relationships between different attributes within a dataset. Conducting this analysis enables the recognition of attribute pairs that exhibit the most significant correlations, offering vital insights into the intrinsic organization of the data. Highly correlated attributes, which indicate the same variance within the dataset, can be further examined to ascertain their significance in model construction. The values of correlation span the spectrum from -1, indicating a perfect inverse linear relationship, to +1, which signifies a perfect direct linear relationship, with 0 representing a lack of correlation. The range in question offers a numerical assessment of the extent of linear correlation between the two variables.

From Fig 1, no features highly related to the target value exist in the dataset. Moreover, some features show negative correlation with target values, while others show positive correlation. Mathematically, the calculation of the Pearson correlation coefficient can be performed according to the subsequent formula:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \quad (1)$$

Here, r represents the Pearson correlation coefficient, x_i denotes the sample's actual value for the x variable, $\bar{x}$ is the average value of the x variable, y_i signifies the actual value of the y variable in the sample, and $\bar{y}$ represents the average value of the y variable.

Fig 2 illustrates the distribution of each attribute within the dataset through histogram visualizations. These histograms delineate the distribution patterns of the respective attributes. Upon examination of these graphical representations, it is evident that the distribution ranges of the attributes vary significantly. Therefore, it should be very beneficial to normalize the input features before entering the data into the machine learning model.

Bar graph of the target category is shown in Fig 3. The dataset is categorically unbalanced.

2.1.2. Data preprocessing. Within the domain of machine learning, the preprocessing of data is recognized as a critical initial step. It covers the original data cleaning and organization operations, and the purpose is to make the data fit the construction and training needs of machine learning models [30]. In practical scenarios, datasets frequently

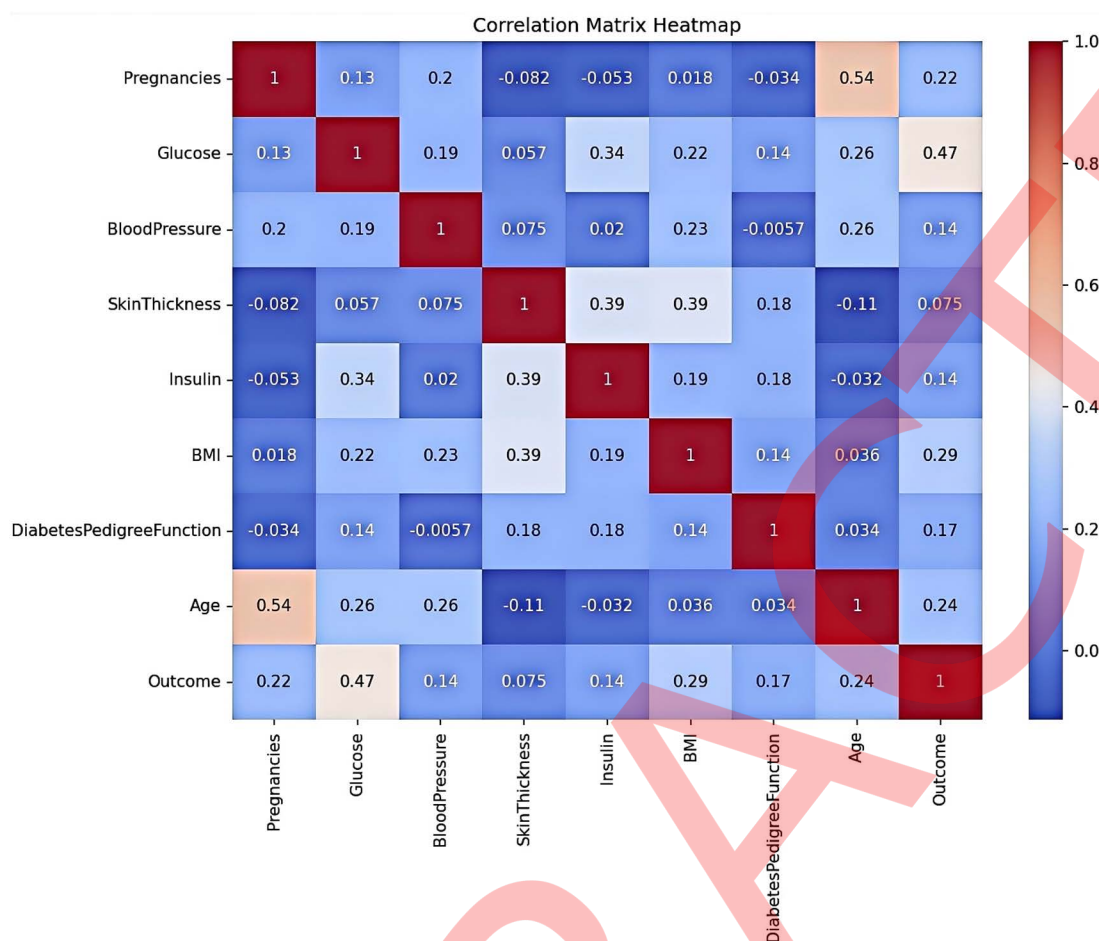


Fig 1. Correlation matrix.

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exhibit incompleteness and inconsistencies, potentially omitting key behavioral patterns or trends, along with numerous inaccuracies. Essentially, data preprocessing serves as a crucial step in the data mining process, transforming raw data into a format that is both readable and comprehensible. This transformation is achieved through various preprocessing techniques. Following an initial data investigation with EDA, the paper followed the established steps to ensure successful data preprocessing.

(1) Identify and address the missing values

In data preprocessing, proper identification and processing of missing values are crucial, otherwise, this study acknowledges the potential for incorrect conclusions due to the presence of missing data in columns 3, 4, and 5. Addressing missing values can be approached through deletion or imputation. Deletion, while straightforward, may not be the most effective method, especially when the dataset is not sufficiently large, as it could introduce further bias. Therefore, it is advisable to employ imputation techniques to maintain data integrity and prevent the exacerbation of biases. Given this, this paper chose the latter replacing the missing values with the mean.

In this scenario, the manuscript presents the computation of the mean values for selected attributes (Blood Pressure, Skin Thickness, and Insulin) to impute missing data. This approach is able to increase the variance of the dataset, effectively avoiding any case of data loss.

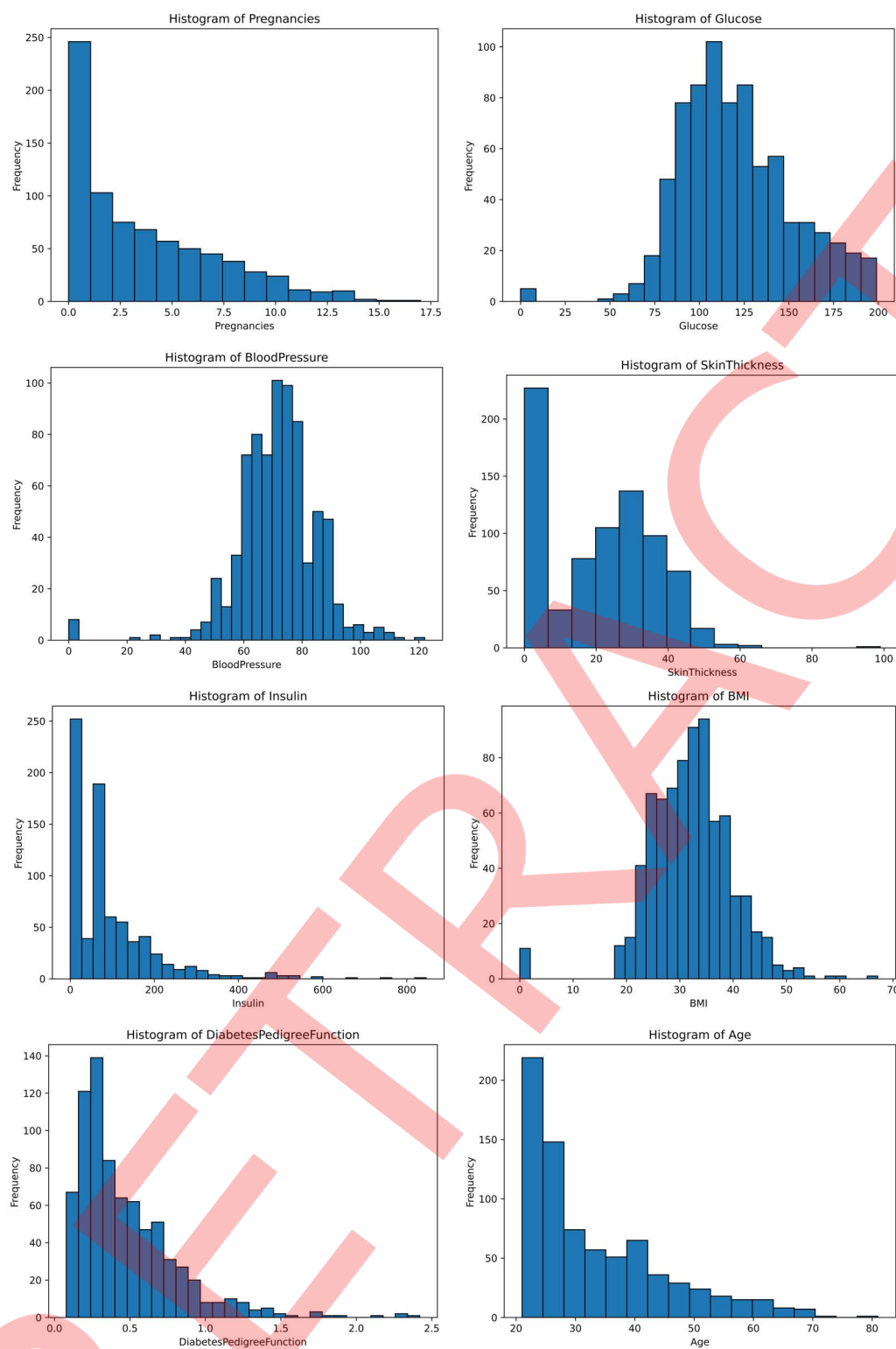


Fig 2. Histogram: Histogram of distribution of Kare public dataset.

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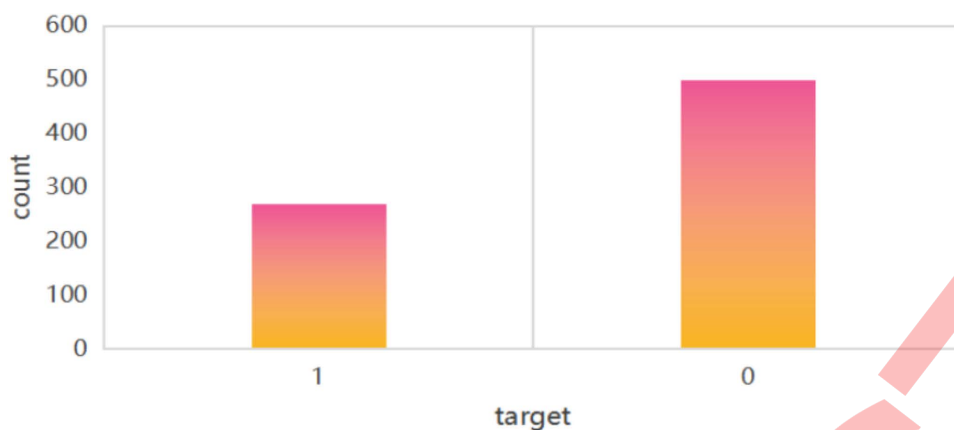


Fig 3. Number of each target category.

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(2) Training and validation sets were established from the dataset.

During the data preprocessing phase, the datasets are usually divided into two independent datasets, where the training dataset is used as the first subset for fitting or training the model, and the test dataset is used as the second subset to validate the fitted model. Commonly, this study employs data partitioning ratios of 70:30 or 80:20, allocating 70% or 80% of the dataset for training the machine learning algorithm, while reserving 30% or 20% for validating the trained model's performance [31]. However, the specific segmentation ratio can be adjusted according to the shape and size of the dataset. In this experiment, the dataset was apportioned such that the training set constituted 80%, while the test set made up the remaining 20%. At the same time, considering the problem of class imbalance in the original dataset, this article introduces the SMOTE (Synthetic Minority Over sampling Technique) algorithm to process the data. The SMOTE algorithm generates new composite samples by interpolating minority class samples, thereby increasing the number of minority class samples and balancing the class distribution of the dataset. After processing with the SMOTE algorithm, a new training set and a new test set were obtained.

(3) Functional scaling

During the data preprocessing stage, feature scaling is applied to normalize the scope of independent variables within a dataset to a predetermined range. Essentially, feature scaling adjusts the distribution of these variables so that they can be effectively compared on an equal footing. This standardization process is crucial for ensuring that the variables are treated consistently and can be accurately analyzed in the context of this paper. The proportion of individual input features may not be the same in the current dataset. For example, "Pregnancies", "Glucose", and "Blood Pressure". In this case, if these features are computed or analyzed, features with a large numerical range may dominate other features, leading to erroneous results. Therefore, this paper must apply functional scaling to eliminate this problem. Gradient descent, a prevalent optimization method for polynomial algorithms, is particularly important for models such as logistic regression and multilayer perceptron (MLP) neural networks. The most discussed method for functional scaling is normalization.

As an important scaling technique, normalization mainly functions to move and rescale the values to ensure that these values end up ultimately [32] between 0 and 1. From a mathematical perspective, the mathematical expression of the normalized equation is:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (2)$$

In this instance, X_{max} and X_{min} represent the upper and lower bounds of the feature's range, respectively.

2.2. System solution method

2.2.1. IRSA. The RSA is an emerging optimization algorithm, which is simple to implement, has strong stability, and few parameters to adjust. In terms of multivariate function solving, it has more reliable convergence with global search ability [33]. IRSA mainly improves RSA in four aspects:

- (1) Population initialization based on Cubic chaos mapping with elite reverse learning

In the execution of the RSA algorithm, the starting points for the individuals are assigned randomly across the search space. As the iterative process proceeds, the best spatial location obtained in each iteration is regarded as an approximate estimate of the global optimal solution, which can be described by the following mathematical model:

$$x_{ij} = \text{rand} \times (UB - LB) + LB, j = 1, 2, \dots, n \quad (3)$$

Here, x_{ij} denotes the j -dimension's value for the i candidate solution; n signifies the dimensionality of the problem at hand. Additionally, rand is a randomly generated number within the interval $[0, 1]$, while LB and UB correspond to the minimum and maximum limits of the problem's feasible region, respectively.

Nevertheless, the initial phase of the RSA algorithm employs a stochastic distribution approach, potentially resulting in a non-uniform spread of individuals across the initial population. This reduction in diversity can adversely affect the algorithm's solution efficiency, and in some cases, may result in optimization search failure [34]. In order to improve the uniformity and ergodicity of the population distribution, we introduce a chaotic sequence for initialization. Different chaotic maps differ in search ability, especially in convergent accuracy. Compared with the logistic chaotic map, the Cubic chaotic map is favored because of its excellent chaotic ergodicity, which has the characteristics of fast optimization and high precision [35]. Based on Cubic chaos mapping, better individuals can be distributed as evenly as possible, and effectively balance the relationship between local optimization and global optimization.

Cubic Chaos mapping (Fig 4) maps the function to $(0,1)$ with the following expression:

$$C_{i+1} = \mu C_i (1 - C_i^2) \quad (4)$$

In the formula, i is the number of populations, $C_i \in (0, 1)$ is the value of the i chaotic mapping function; Taking the chaotic parameters of $C_0 = 0.3$ and $\mu = 2.595$, the Cubic chaotic map has good traversal between $(0, 1)$ [36].

With the initial population generated by the Cubic chaos mapping function, then the initialization formula of RSA is changed to:

$$x_{ij} = C_i \times (UB - LB) + LB, j = 1, 2, \dots, n \quad (5)$$

Furthermore, the elite reverse learning strategy is applied to augment the diversity of the initial population. This approach creates an inverse population [37] starting from a randomly generated one. The population with the highest fitness function value, when compared with the original and the inverted populations, is chosen as the starting point for subsequent iterative processes. The mathematical expression for the inverse population is presented below:

$$X'_i = \text{rand}(LB + UB) - X_i \quad (6)$$

X_i denotes the position vector for the original population, while X'_i represents the position vector of the corresponding inverted population.

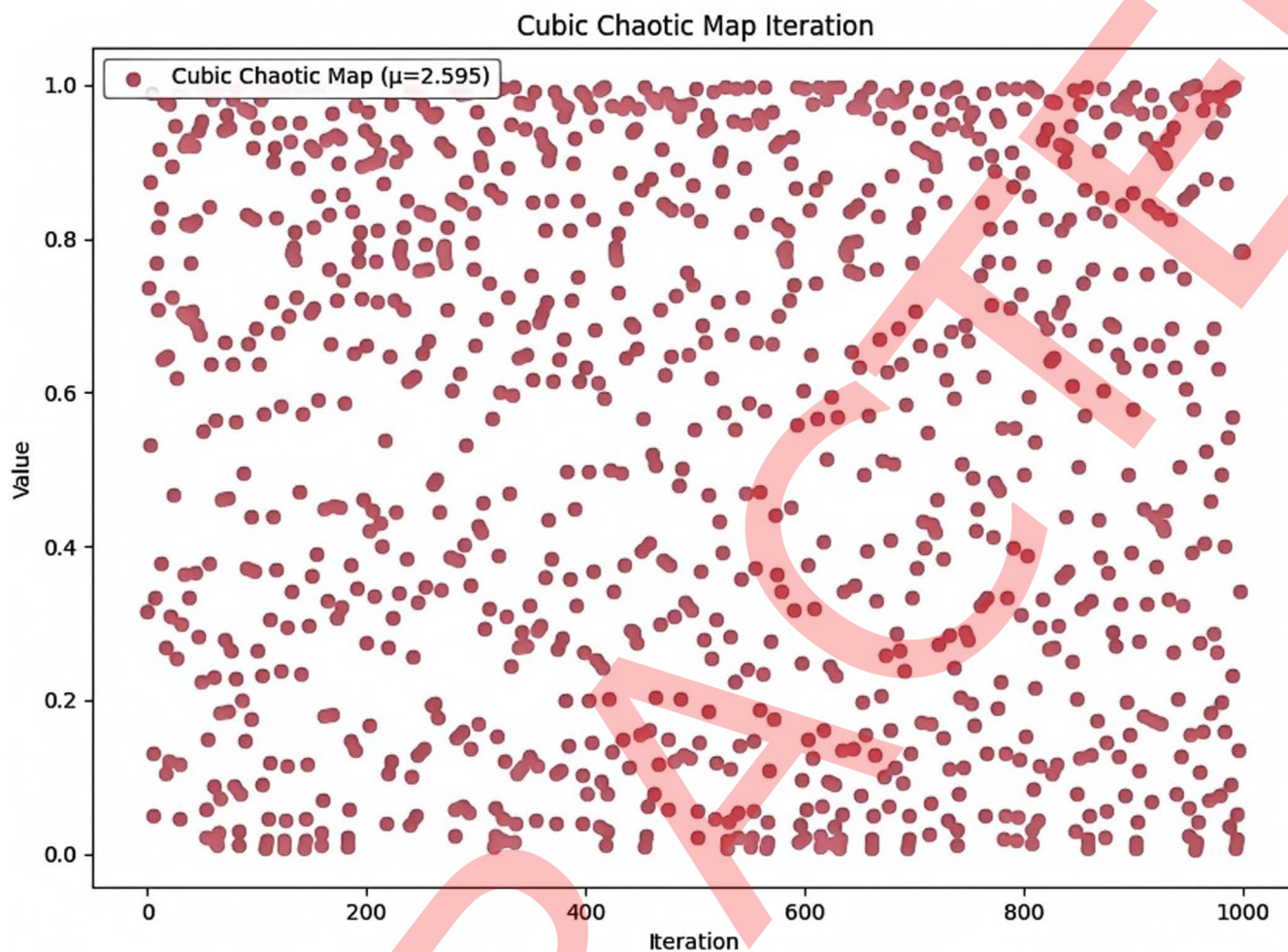


Fig 4. An iterative plot of the Cubic chaotic map.

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Change the mathematical formula for generating the reverse population in IRSA by combining the elite reverse learning strategy with Cubic chaotic mapping to:

$$X'_i = C_i(LB + UB) - X_i \quad (7)$$

(2) Introduction of the golden ratio algorithm in the surround search phase

In the RSA algorithm, crocodiles utilize two distinct strategies during the encirclement phase, reflective of their predatory behaviors: the high-altitude traversal tactic and the ventral surface movement strategy. Due to potential interference factors in the environment, these two strategies will to some extent limit the direct path for crocodiles to approach their prey. However, this restriction actually promotes their exploration of a wider area, which not only increases the possibility of discovering prey gathering places, but also provides valuable spatial information for future hunting activities. The exploration phase is contingent upon two distinct conditions. The application of the high-altitude walking strategy is governed by the condition $t \leq \frac{T}{4}$, whereas the abdominal walking strategy is invoked under the condition $\frac{T}{4} < t \leq \frac{T}{2}$. This indicates that

approximately half of the exploratory iterations will fulfill the criteria for high-altitude walking, with the remaining half designated for abdominal walking. High-altitude walking and abdominal walking constitute distinct approaches to exploration and searching. The positional update during the exploration phase can be articulated by the following equation:

$$x_{(ij)}(t+1) = \begin{cases} Best_j(t) - \eta_{(ij)}(t) \times \beta - R_{(ij)}(t) \times rand, t \leq \frac{T}{4} \\ Best_j \times x_{(r_1,j)} \times ES(t) \times rand, t > \frac{T}{4} \end{cases} \quad (8)$$

Here, $Best_j$ denotes the j component of the current best solution; $rand$ is a randomly generated number with in the range $(0, 1)$; t signifies the ongoing iteration index; T is the predefined limit on the number of iterations; $\eta_{(ij)}$ is the hunting mechanism for the j -dimension of the i candidate solution. The calculation formula is as follows:

$$\eta_{(ij)} = Best_j(t) \times P_{(ij)} \quad (9)$$

β , a sensitive parameter, dictates the precision of exploration (resembling high-altitude walking) in the enveloping phase of the iterative process, with a fixed value of 0.1. The reduction function $R_{(ij)}(t)$ is a value used to reduce the search area, and the calculation formula is:

$$R_{(ij)} = \frac{Best_j(t) - x_{(r_2,j)}}{Best_j(t) + \varepsilon} \quad (10)$$

r_1 denotes a randomly generated integer within the range $[1, N]$; $x_{(r_1,j)}$ signifies the position of the j -dimension for the r_1 randomly selected candidate solution. N denotes the overall number of candidate solutions. The evolution factor, $ES(t)$, which is a stochastic ratio, varies randomly within the range of -2 – 2 during the iterative process. The calculation formula is:

$$ES(t) = 2 \times r_3 \times \left(1 - \frac{t}{T}\right) \quad (11)$$

Among them, ε is an exceedingly small positive value; r_2 is a randomly selected integer from the interval $[1, N]$; r_3 denotes a straightforward random integer within the range $[-1, 1]$. The j -dimensional position's relative percentage deviation from the optimal solution to the current solution is signified by $P_{(ij)}$, according to the formula that follows:

$$P_{(ij)} = \alpha + \frac{x_{(ij)} - M(x_i)}{Best_j(t) \times (UB_{(j)} - LB_{(j)}) + \varepsilon} \quad (12)$$

Among them, $M(x_i)$ signifies the mean location of the i candidate solution and its calculation formula is:

$$M(x_i) = \frac{1}{n} \sum_{j=1}^n x_{(ij)} \quad (13)$$

$UB_{(j)}$ and $LB_{(j)}$ denote the upper and lower limits, respectively, for the position in the j -dimensional space. The parameter α , which is pivotal for regulating the accuracy of the search within the hunting cooperation iterations (i.e., the disparity between potential solutions), is set to a constant value of 0.1 for this research.

In this study, we introduce an advanced version of the RSA, known as the IRSA, which addresses common limitations of the original algorithm, including its tendency to converge slowly and get trapped in local optima. This refinement integrates the golden sine technique to enhance the local search process of the RSA. The innovative approach for position updating is described as follows:

$$x_{(ij)}(t+1) = \begin{cases} a \times Best_j(t) - (1-a)\eta_{(ij)}(t) \times \beta - R_{(ij)}(t) \times rand, t \leq \frac{T}{4} \\ a \times Best_j \times x_{(r1,j)} \times ES(t) \times rand, \frac{T}{4} < t \leq \frac{T}{2} \end{cases} \quad (14)$$

The Golden Sine Algorithm leverages the mathematical sine function for iterative optimization processes, offering substantial benefits such as rapid convergence, enhanced robustness, straightforward implementation, and a minimal number of tunable parameters and operators, as referenced in [38]. It introduces the golden ratio in the wrapper search phase of the RSA algorithm, which is a fixed value of $(\sqrt{5}-1)/2$. In the process of algorithm iteration, when entering the later stage and the optimization process is basically completed, in order to avoid excessive reliance on the discovered best solution $Best_j$, the golden ratio algorithm is adopted. Perform golden ratio processing on $Best_j$ solutions, multiply them by the golden ratio of a , in order to achieve a balance between exploration and development. This approach ensures that the algorithm effectively explores new potential regions while fully utilizing the known optimal region, which helps improve the global search capability of IRSA and prevents premature convergence to local optimal solutions.

(3) The hunting phase with the introduction of non-linear adjustment factors

Similar to the encirclement phase, crocodiles employ two distinct strategies during their hunting phase: coordinated hunting and cooperative hunting, which serve to refine local exploration. Unlike the encirclement stage, these two strategies are beneficial for crocodiles to approach their prey. During the hunting phase of the RSA algorithm, the search space is effectively utilized to identify the optimal solution. The success is mainly due to two key strategies: the coordinated hunting method and the cooperative hunting method. The mathematical model is as follows:

$$x_{(ij)}(t+1) = \begin{cases} Best_j(t) \times P_{(ij)}(t) \times rand, \frac{T}{4} < t < \frac{3T}{4} \\ Best_j(t) - \eta_{(ij)}(t) \times \varepsilon - R_{(ij)}(t) \times rand, \frac{3T}{4} < t < T \end{cases} \quad (15)$$

To bolster the algorithm's capacity for global exploration throughout the optimization process, IRSA introduces a non-linear adjustment factor of A to improve the hunting phase of RSA. The improvement results are as follows:

$$x_{(ij)}(t+1) = \begin{cases} Best_j(t) \times A \times P_{(ij)}(t) \times rand, \frac{T}{4} < t < \frac{3T}{4} \\ Best_j(t) - \eta_{(ij)}(t) \times \varepsilon - R_{(ij)}(t) \times rand, \frac{3T}{4} < t < T \end{cases} \quad (16)$$

The formula for nonlinear adjustment factor A is described as follows:

$$A = \arctan \left[e^{\left(-\frac{Iter}{MAXIter} \right)^3} \right] \quad (17)$$

Introduced as a decrementing composite function, this factor is acutely responsive to the algorithm's dynamic tuning, enabling it to modulate the search velocity in accordance with the iteration progression. Here, $Iter$ denotes the ongoing iteration index, while $MaxIter$ signifies the threshold for the total iterations. As iterations unfold, parameter A undergoes a gradual decline. This attrition facilitates a reduced reliance on previously identified optimal solutions, thereby propelling the algorithm to pursue uncharted potential areas in the advanced stages of iteration. This strategy effectively circumvents premature convergence, enhancing the algorithm's extensive search efficacy.

(4) Introducing the **sigmoid ()** function

In the mathematical expression of the reduction function $R_{(ij)}(t)$ its denominator contains a very small positive number. The **sigmoid()** function is introduced. The improved mathematical expression is as follows:

$$R(i,j) = \frac{Best_j(t) - x(r2,j)}{Best_j(t) + \xi * \frac{1}{1+e^x}} \quad (18)$$

Taking the ratio of the current iteration count to the total iteration count as input, an optimization algorithm can exhibit a dynamically changing characteristic over time. And it also allows IRSA to flexibly adjust its search strategy at different iteration stages. In the initial stage of algorithm operation, a larger proportion of iterations is inputted into the **sigmoid ()** function, which can prompt the algorithm to conduct a more extensive global search, greatly increasing the probability of finding a better solution; In the later stage of optimization, as the proportion of iterations gradually decreases, the output of the **sigmoid ()** function also tends to stabilize. At this time, the algorithm is more inclined to perform local fine search, which is extremely beneficial for improving the accuracy of the solution and accelerating the convergence speed.

2.2.2. BP. Representing a multi-layer perceptron, the BP neural network comprises an input layer, numerous hidden layers, and an output layer [39]. It is trained through the backpropagation algorithm, aiming to maximize the approximation of the network's predicted output to the true value. Introduced by Rumelhart and McClelland in 1986, the BP neural network has become a prevalent model within the realm of deep learning. This architecture is characterized by its input layer, followed by a series of hidden layers, culminating in an output layer. This network structure is designed in such a way that information can be transmitted layer by layer from the input layer to the output layer. Neurons in each layer are capable of processing the output of the previous layer and creating new feature representations. Fig 5 shows a typical BP neural network architecture, where the connections between layers reflect the depth characteristics of the network, providing a foundation for complex function mapping.

As shown in Fig 5, the input layer takes in external input data and transmits it to the hidden layer. The hidden layer may consist of multiple layers, with each layer containing multiple neurons. Neurons are the fundamental computing units of neural networks. They carry out weighted summation on the input data and conduct nonlinear transformations via activation functions to yield output values. The output layer determines the final prediction result on the basis of the output from the hidden layer.

The training protocol for the BP neural network entails adjusting the weights and biases through an error backpropagation mechanism, targeting the reduction of the prediction error compared to the expected outcomes. The training protocol is segmented into two distinct phases:

(1) Positive propagation

During the feedforward phase, data is gradually refined by each successive layer. It starts from the input layer, passes through the hidden layers, and finally results in the output layer producing outputs.

$$h_i = f\left(\sum_{j=1}^n w_{ij}x_j + b_i\right) \quad (19)$$

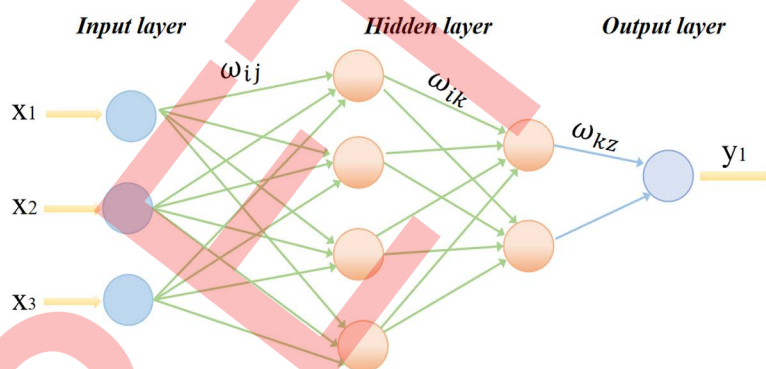


Fig 5. BP neural network architecture diagram.

<https://doi.org/10.1371/journal.pone.0324759.g005>

Within this framework, x_j denotes the output of the j neuron in the input layer, w_{ij} represents the synaptic weight connecting the input to the hidden layer, b_i signifies the threshold value of the neuron in the hidden layer, and $f(\cdot)$ corresponds to the activation function. In a canonical BP neural network setup, neurons within the hidden layer typically employ the same activation function to maintain uniformity and stability during information processing. In contrast, the output layer may apply a unique activation function, which is frequently customized based on the model's aims or the nature of the predictive task. The core function of an activation function is to add non-linearity to the network, enabling it to model sophisticated nonlinear relationships between input and output variables. As a result, BP neural networks can handle a wider range of data patterns and functional mapping issues. During the feedforward process, data is processed systematically layer by layer, starting from the input layer, passing through the hidden layers, and finally leading to the output of the output layer.

Activation functions that are frequently utilized encompass the Sigmoid and ReLU functions [40]. The Sigmoid function is a classic activation function with a mathematical expression of $f(x) = \frac{1}{1+e^{-x}}$. As shown in Fig 6. Its output range is between (0,1), making it an ideal choice for binary classification problems [41]. The output value of the Sigmoid function can be interpreted as probability, which makes it very popular in early neural network research. The function known as ReLU (Rectified Linear Unit) is defined as follows $f(x) = \max(0, x)$. The ReLU function has a constant gradient of 1 in the

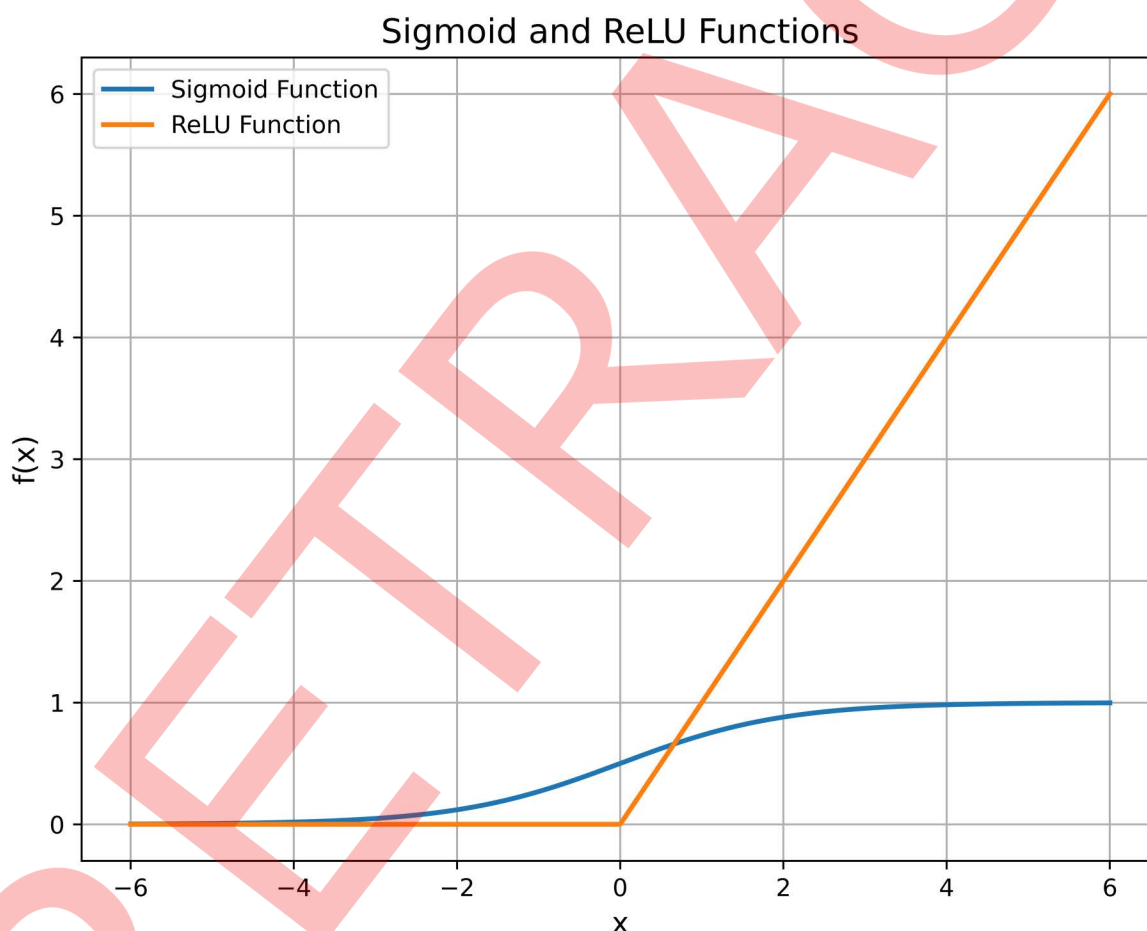


Fig 6. Activation functions: Sigmoid function and ReLU function.

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positive interval, which ensures that the gradient does not disappear during training, thereby accelerating the convergence speed [42]. In addition, the ReLU function has low computational complexity and can effectively reduce the computational burden of the model.

(2) Backpropagation

In the backpropagation stage, the error of the output layer is first calculated, and then the error is backpropagated to the hidden layer, adjusting the weights and thresholds layer by layer. Typically, error computation makes use of loss metrics like Mean Squared Error (MSE) or Cross Entropy.

The calculation method for the error of neurons in the output layer is:

$$\delta_i = (y_i - \hat{y}_i) \cdot f'(h_i) \quad (20)$$

Among them, y_i is the target output, $\hat{y}_i$ is the actual output of the network, and $f'(\cdot)$ is the derivative of the activation function.

The calculation method for the error of hidden layer neurons is:

$$\delta_j = \left(\sum_{i=1}^m \delta_i w_{ij} \right) \cdot f'(h_j) \quad (21)$$

Among them, m represents the number of neurons in the output layer.

The formula for adjusting weights and thresholds based on the calculation of errors is:

$$w_{ij} = w_{ij} - \eta \delta_i h_j \quad (22)$$

$$b_i = b_i - \eta \delta_i \quad (23)$$

Among them, η is the learning rate, used to control the adjustment step size.

By continuously repeating the process of forward and backward propagation, the BP neural network gradually learns the mapping relationship between input data and target output, thereby achieving prediction of unknown data.

2.2.3. Training BP with IRSA. This section evaluates the utilization of the IRSA in enhancing the training process of a BP neural network for diabetes prediction. Fig 7 depicts the schematic representation of the methodology in question. The IRSA is designed to navigate extensive solution spaces. Within the framework of BP network training, this research employs the fitness function as the benchmark for optimization, meticulously adjusting the network's weights and biases. The objective is to pinpoint the most effective weights and biases configuration that minimizes the fitness value, consequently improving the network's predictive precision for diabetes data.

In the IRSA framework, a problem's resolution is encapsulated within a solution vector, the dimensions of which are dictated by the neural network's weight configurations and the aggregate count of thresholds. Each component of the solution vector corresponds to the weight or bias at a distinct location within the neural network architecture. A population using a potential solution called "population" is formed by multiple solution vectors. When using IRSA to optimize BP, the position of each individual in the population is a vector representing the weight and bias value of BP. During the iteration process of the IRSA algorithm, individuals in the population are continuously updated to gradually approach the optimal solution. For example, if the publicly available dataset for Khare is instantiated as a BP network with 8 input layer nodes, 8 hidden nodes, and 2 output nodes (the number of output categories), then the dimension of each individual position in a

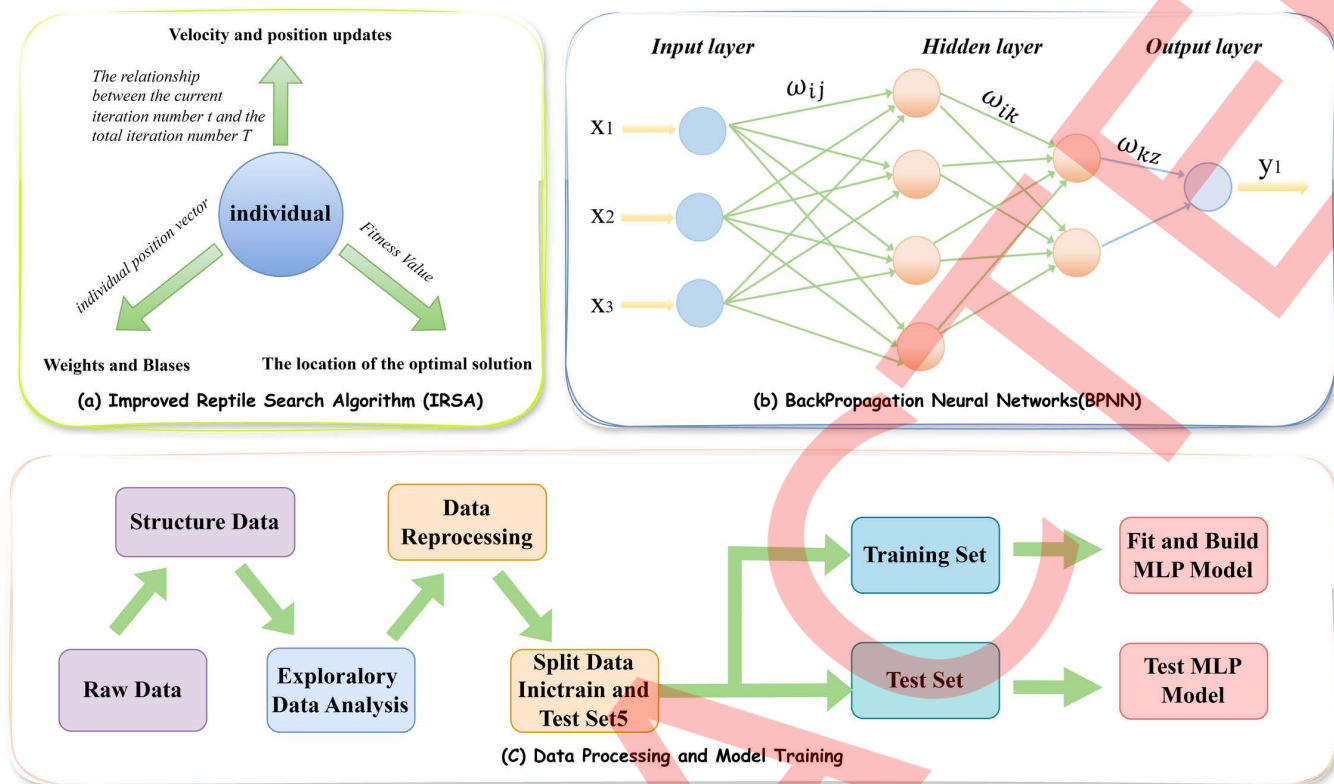


Fig 7. Flow chart of the proposed IRSA-BP diabetes prediction model. This paper separately benchmarks the performance of the algorithm by using the diabetes dataset provided by Khare and the diabetes dataset processed with the SMOTE algorithm. Before model training, detailed preprocessing was performed on the data, including identification and processing of missing values, transformation of categorical variables, and standardized scaling of features, to ensure data quality and provide a solid foundation for accurate evaluation of algorithm performance.

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fully connected BP neural network population is $(8 \times 8) + 8 + (8 \times 2) + 2 = 94$, which is the total number of weights and bias values.

2.3. Comparative optimization algorithms

This study compared three ML optimized BP neural networks with the IRSA-BP proposed in this paper.

2.3.1. Reptile search algorithm. The Reptile Search Algorithm (RSA) emulates the hunting strategies of crocodiles, functioning as an optimization technique that seeks the optimal solution through a two-phase simulation of entrapment and pursuit. Compared with IRSA-BP, RSA has a simpler overall initialization method, namely random initialization. In some cases, this may result in slower convergence speeds. In its search strategy, RSA's two-stage hunting strategy is not as flexible as IRSA's improved strategy.

2.3.2. Skyhawk optimization algorithm. The Skyhawk Optimization Algorithm (AO) is a metaheuristic optimization algorithm that simulates the hunting behavior of eagles, proposed by Abualigah et al. in 2021. This algorithm simulates the different strategies of eagles during hunting, such as high-altitude soaring, short gliding attacks, low altitude slow descent attacks, and walking to grab prey, in order to find the optimal solution [43]. AO may be more suitable for problems with relatively simple search space, while for complex problems such as prediction, more comprehensive optimization methods of IRSA-BP can provide better prediction.

2.3.3. Marine predator algorithm. The Marine Predator Algorithm (MPA) draws inspiration from the hunting patterns of marine hunters, particularly leveraging their Lévy flights and Brownian motions, along with the strategy for maximizing the encounter rate with prey during their foraging activities. MPA simulates these natural behaviors to find the optimal solution, and adopts different strategies at different iteration stages to balance global search and local search, in order to improve the algorithm's optimization ability [44]. However, compared to IRSA-BP, MPA is not yet effective in fine-tuning the weights and biases of BP neural networks.

3. Experiment and results

This study aims to explore the difference between IRSA and traditional ML optimization algorithms in training BP neural networks, and further evaluate the application potential of these methods in developing binary classification models for diabetes diagnosis.

3.1. Experimental setup

All research experiments were conducted using MATLAB [45], relying on functions from MATLAB toolboxes such as cvpartition [46], mapminmax [47], ind2vec [48], confusionchar [49], statsofMeasure [50], etc. Table 2 summarizes the control parameters of the IRSA algorithm used for training BP networks.

The objective function that BP network needs to minimize is the fitness function. Fitness is the most basic and widely used performance evaluation indicator, and its calculation formula is as follows:

$$f = 1 - \frac{\sum_{i=1}^n I(\hat{y}_i = y_i)}{n} \quad (24)$$

Among them, the training set sample size is n , the neural network prediction value is $\hat{y}_i$, the true value is y_i , an $I(\hat{y}_i = y_i)$ is the indicator function. When $\hat{y}_i = y_i$, $I(\hat{y}_i = y_i) = 1$; When $\hat{y}_i \neq y_i$ is $I(\hat{y}_i = y_i) = 0$.

The population size is configured at 10, representing a balance point. While expanding the population can offer a broader range of search trajectories, potentially elevating the likelihood of encountering the global optimum, it does not invariably ensure accelerated convergence. This is attributed to the fact that an increased individual count correlates with longer computational times per epoch, potentially diminishing the frequency of weight adjustments that can be accomplished within the constraints of available training time. Therefore, choosing a moderate population size is crucial as it can strike a balance between search efficiency and computational cost, thereby optimizing the use of computing resources while ensuring algorithm performance.

The operation is deemed successful and halted when either the iteration count reaches its maximum of 8 or the optimal fitness value falls within the preset error tolerance. The proposed BP neural network, which incorporates an optimized RSA algorithm for training, comprises 8 hidden neurons. This network was subsequently benchmarked against three other ML optimized BP neural networks to assess and evaluate the performance of the proposed IRSA-BP model.

Table 2. Control Parameters of IRSA Algorithm.

Parameters	Size
Population size	10
Number of iterations	8
Number of training epochs	10000
Upper bound of search range	3
Lower bound of search range	-3
Learning rate	0.01
Number of hidden layer nodes	8

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3.2. Evaluation indicators

For this investigation, the dataset was segmented into 80% for training the models and 20% for validation. A detailed evaluation of model efficacy was performed, using a set of well-recognized classification metrics to ascertain the precision of the model's predictive capacity. The examination concentrated on metrics such as accuracy, precision, recall, and the F1 score, as detailed in [51,52]. These metrics stem from the confusion matrix, which consists of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). Furthermore, to appraise the model's classification proficiency, the analysis included the examination of the ROC curve and the computation of the AUC. The evaluation criteria included accuracy, recall, the F1 score, and the AUC-ROC. The accuracy was determined by the proportion of instances correctly classified relative to the total number of samples. This is illustrated in Equation 25:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (25)$$

Precision is calculated as the proportion of true positive cases relative to all cases that are labeled as positive, as illustrated in Equation 26:

$$Precision = \frac{TP}{TP + FP} \quad (26)$$

Recall is defined as the ratio of true positive samples that the model successfully identifies as positive out of all actual positive cases. As shown in Equation 27:

$$Recall = \frac{TP}{TP + FN} \quad (27)$$

Equation 28 specifies the F1 score, which is the harmonic mean of precision and recall.

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (28)$$

The performance of binary classification algorithms can be graphically evaluated using the ROC curve. This curve plots the true positive rate (TPR), or recall rate, against the false positive rate (FPR) across various threshold settings. The AUC, representing the area under the ROC curve, is an indicator derived from this visual representation. The FPR can be calculated using the following formula:

$$FPR = \frac{FP}{TN + FP} \quad (29)$$

TP represents the count of instances where the model accurately identifies the presence of diabetes, whereas TN corresponds to the cases where the model correctly ascertains the absence of the condition. On the other hand, FP refers to the scenarios where the model mistakenly classifies individuals as diabetic when they are not, and FN indicates the cases where the model erroneously misses predicting diabetes in those who are actually affected.

3.3. Results and discussion

In this study, this paper explored a diagnosis method of diabetes using machine learning technology, and trained and evaluated various models including the proposed IRSA-BP model. Comprehensive model performance evaluations were

Table 3. Comparison of Testing Performance Using Various Performance Evaluation Indicators.

Algorithm	Accuracy Rate	AUC	Precision Rate	Recall Rate	F1 - score
IRSA-BP	0.836	0.878	0.810	0.701	0.752
RSA-BP	0.796	0.852	0.759	0.613	0.678
MPA-BP	0.798	0.868	0.726	0.679	0.701
AO-BP	0.813	0.862	0.787	0.641	0.708

<https://doi.org/10.1371/journal.pone.0324759.t003>

Table 4. Comparison of testing performance of various performance evaluation indicators in category balanced datasets.

Algorithm	Accuracy Rate	AUC	Precision Rate	Recall Rate	F1 - score
IRSA-BP	0.790	0.845	0.761	0.851	0.803
RSA-BP	0.730	0.803	0.708	0.792	0.748
MPA-BP	0.724	0.723	0.731	0.812	0.767
AO-BP	0.720	0.798	0.714	0.743	0.728

<https://doi.org/10.1371/journal.pone.0324759.t004>

conducted on both the publicly available dataset of Khare and the dataset processed using the SMOTE algorithm, including accuracy AUC, Accuracy, recall, and F1 score. The outcomes of these experiments are presented in [Table 3](#) and [Table 4](#).

From the experimental results, it is clear that these predictive features play an important role in the model. By adjusting the analysis of different features, the model was able to accurately capture the signs of diabetes onset. For example, the feature Glucose has a high weighting in the model, which is in line with its key position in the diagnosis of diabetes. Higher blood glucose levels often imply an increased risk of diabetes, and the model is able to accurately predict whether or not an individual has diabetes by effectively capturing the Glucose feature. Similarly, other features such as the number of pregnancies and blood pressure positively influence the model's predictions, which together improve the accuracy and reliability of the model.

In this paper, when comparing the computational efficiency, it was found that by comparing the running time of each algorithm during training it is found that the IRSA-BP algorithm is able to find the optimal solution in fewer iterations. In terms of performance, this paper uses a variety of classification metrics to evaluate the algorithms. As shown in [Figs 8](#) and [9](#), the indicator curves of the IRSA-BP algorithm are above the other three algorithms, indicating that the IRSA-BP model outperforms the other three comparison algorithms in all of these indicators. These results indicate that the IRSA-BP algorithm not only has high diabetes prediction accuracy but also high computational efficiency, providing an accurate and



Fig 8. Curve chart of various algorithm indicators before category balance.

<https://doi.org/10.1371/journal.pone.0324759.g008>

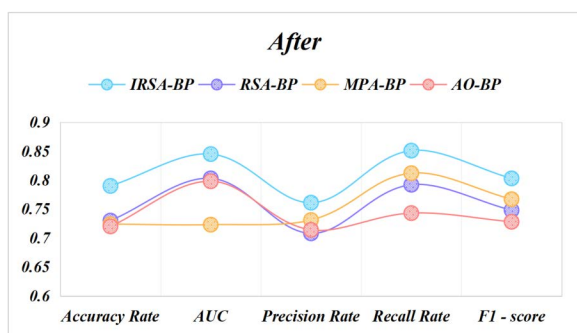


Fig 9. Curve graph of various algorithm indicators after category balance.

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Fig 10. Bar chart comparing various algorithm indicators before and after category balance.

<https://doi.org/10.1371/journal.pone.0324759.g010>

reliable machine learning method for diabetes diagnosis. For further analysis, Fig 10 provides a bar chart comparing the performance indicators of various algorithms before and after category balancing, which also demonstrates the superior performance of the IRSA-BP algorithm.

Although the IRSA – BP model shows high accuracy, this paper acknowledges the importance of interpretability in medical diagnosis. To balance interpretability and accuracy, this paper adopted multiple strategies. This paper carefully

designed the BP neural network with 8 hidden neurons to simplify the information flow and enhance interpretability without major accuracy loss. The dataset used has 8 well-defined, clinically meaningful medical attributes related to diabetes, ensuring the model's predictions are based on interpretable factors.

4. Conclusion and prospect

A predictive model for diabetes, which integrates an IRSA to optimize the BP neural network, is presented in this paper. And the methodology used makes several key contributions.

In terms of algorithmic innovation, the proposed IRSA effectively enhances the global search capability and convergence efficiency of traditional RSA through multiple improvements, such as Cubic chaos mapping initialization with elite reverse learning, golden ratio-enhanced local search, nonlinear adjustment factors, and sigmoid-based dynamic adjustment. In terms of performance, the experimental data indicate that the IRSA-BP model outperforms other comparative algorithms in critical performance metrics for diabetes prediction, with an accuracy of 83.6% before category balance and 79% after category balance, demonstrating its potential for handling complex medical data.

Regarding data preprocessing, this study rigorously analyzed and preprocessed the Khare dataset, specifically focused on female subjects aged 21 and older. While this specialized dataset allowed an effective demonstration of the model's predictive potential, it also revealed the necessity of extending research to broader populations, encompassing different genders, ages, and ethnicities.

In future research, will explore several avenues to advance the IRSA-BP methodology and its applicability Firstly, extensive validation and analysis of IRSA-BP across diverse datasets, including different populations and regions, will be conducted to enhance its generalizability. Secondly, the potential of IRSA to optimize other neural network architectures [53], such as recurrent neural networks (RNNs) for medical time-series data analysis, will be explored to further expand its application scope.

Regarding the IRSA – BP model, it is planned to study several components in more depth. For the IRSA algorithm, the impact of different chaotic mapping functions in the overall initialization phase will be investigated. Although Cubic chaotic map is used in this study, other chaotic mappings may have different advantages in terms of population diversity and convergence speed. Secondly, the nonlinear tuning factors of the parameters in the Golden Ratio algorithm will also be investigated. Some parameters will be fine-tuned to further enhance the global and local search capabilities of the IRSA algorithm, thus providing better optimization results for BP neural networks.

In the IRSA-BP neural network section, the optimal number of hidden layers and neurons will be explored. The current model uses 8 hidden neurons, but this may not be the optimal configuration for all datasets and prediction tasks. By adjusting these structural parameters, the aim is to improve the network's ability to capture complex relationships in healthcare data. These studies are expected to further deepen the understanding of the IRSA – BP model and promote its application and development in the field of medical diagnosis.

Author contributions

Writing – original draft: Wan-Hua Zhang, Zi-Xun Zhang.

Writing – review & editing: Wan-Hua Zhang, Zi-Xun Zhang.

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